

Math 290

Practice Quiz 2 — Version E

(extra practice, Sections 6–11)

1. Short response.

(a) Evaluate the following proposed theorem and proof.

Theorem: Let $x \in \mathbb{Z}$. If $2 \mid x$, then $2 \nmid (x^2 + 1)$.

Proof. Assume, by way of contradiction, that $2 \mid x$ and $2 \mid (x^2 + 1)$ for some $x \in \mathbb{Z}$. We can then write $x = 2y$ and $x^2 + 1 = 2z$ for some $y, z \in \mathbb{Z}$. Substituting the first equality into the second, we get $4y^2 + 1 = 2z$. Therefore

$$1 = 2z - 4y^2 = 2(z - 2y^2)$$

is even, which is false. Thus the original implication is true. $\square$

- A. The theorem is true and the proof is correct.
- B. The theorem is false.
- C. The theorem is true, but the proof is incorrect because a proof by contradiction should assume $2 \nmid x$.
- D. The theorem is true, but the proof is incorrect because it should only assume $2 \mid (x^2 + 1)$.
- E. The theorem is true, but the proof is incorrect because it only proves the statement for one value of x .

(b) True or false: There exist $a, b \in \mathbb{N}$ such that $a^2 \mid b^3$ but $a \nmid b$.

(c) True or false: Given $a, b \in \mathbb{Z}$, if $a \mid b$ and $b \mid a$, then $a = b$.

2. Let $a, b, c, d, n \in \mathbb{Z}$. Prove that if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{n}$. Then use this result to decide whether $19^3 \equiv 5^3 \pmod{7}$.

3. Prove or disprove each of the following statements.

(a) There exists an integer x such that $x^2 + x$ is odd.

(b) For each $s \in \mathbb{Z}$, if $6s - 3$ is odd, then s is odd.

4. Let A , B , and C be sets.

(a) Prove or disprove: If $A \cup C \subseteq B \cup C$, then $A \subseteq B$.

(b) Prove the following statement (giving a *single* justification for every step you make). If $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$, then $A \subseteq B$.